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(9610) BL02
Report on the examination

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REPORT ON EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BL02 JANUARY 2024

It appeared that most students were able to fully complete the paper and attempt the majority of the questions. The questions on this paper discriminated well, with 21 of the questions having discrimination indices greater than 0.4. (The discrimination index is a measure of correlation and indicates the extent to which an item discriminates between high-attaining and low-attaining students). The paper produced a range of marks from 0 to 67 (out of 75) and the mean mark of 33.5 was slightly lower than that of BL02 in the summer of 2023 (mean mark 36.2). Students demonstrated good knowledge of subject areas such as the structure of HIV and cell structure, eg in explaining how features of a plasma cell allow for large numbers of antibodies to be produced quickly. Less success was seen with the understanding gained from the required practical on measuring the rate of water uptake using a potometer. There were four calculation questions in the paper; two of these were answered well (**05.3** and **06.5**) and two not so well (**04.2** and especially **01.2**).

QUESTION 01

Question **01.1** was a straightforward question requiring students to name the nucleic acid and an enzyme found within the capsid of HIV. The question was answered well with just under 75% of students gaining full marks.

Students found the calculation in **01.2** more challenging - only around 20% of responses were given the full two marks. Various errors were seen, but the most common was in answers that did not make use of the scale bar. It was also common to see errors in converting between units and not following the instruction to give the final answer to 2 significant figures.

In question, **01.3** students had to describe how viral proteins are produced inside an infected helper T cell. The question discriminated well, with around 65% of students gaining at least one mark and just over 10% of students getting full marks. Many students correctly described viral RNA being used a template to form cDNA but did not state the enzyme involved in the process. There were many vague descriptions of how viral proteins were produced, without any mention of viral mRNA or the fact that this would be translated at the ribosomes. Very few students referred to the role of protease in modifying the viral proteins.

01.4 was an application question where students had to recognise that the drug AZT is similar in shape to thymidine. Around 60% of students were able to do this and correctly described that AZT could compete with thymidine or that it could be used as nucleotide. Very few students were able to expand any more on this and link to how the drug could prevent the onset of AIDS.

QUESTION 02

In this question students had to design an investigation into the effect of exercise on heart rate. **02.1** was answered well, with over 75% of students gaining at least two marks and just over 15% gaining full marks. The majority of students described the need to take measurements of heart rate before and after exercise (as a minimum) and many suggested suitable safety precautions for the investigation. It was quite common for students to miss the idea of taking repeat measurements to enable the calculation of mean values. In some responses, the actual detail of the exercise was unclear and not always given with a suitable control variable.

In questions **02.2**, just over 25% of students gained both marking points, with close to 45% gaining one mark. A number of students misinterpreted the question and gave a sketch graph of the expected results, rather than a results table. It was also common for the units to be missing from the column headings.

Question **02.3** was answered less well. Over 65% of students gained at least one mark but only around 7% of students gained all three marks. Many students correctly named a suitable statistical test such as the t-test but few could give a correct reason for choosing the test and even fewer could describe the outcome of the test if exercise had a significant effect.

In question **02.4** students had to give two factors, other than exercise, that affect heart rate. The question was answered well with around 40% of students gaining one mark and 50% of students gaining full marks. There was a range of suitable ideas but the most commonly seen factors were stress, consumption of caffeine and the general health condition of the participants.

QUESTION 03

Question **03.1** was answered very well with just under 75% of students gaining the two marks available. Many students gave a combination of ideas linked to either the environmental factors or aspects related to the aphids or crop plants used in the investigation. Answers concerning the mass or size of the seeds were not considered to be suitable control variables.

In question **03.2**, only around 30% of students gained full marks. There were various incorrect answers such as just allowing more time for the water to evaporate. Some suggested pressing the plant samples with tissue paper to remove water or to test for transpiration to check if water was still inside.

In question **03.3**, many students correctly identified that the individual effect of both the fungal pathogen and the aphids caused a decrease in dry mass of the plants. For this reason, over 75% of students gained at least one mark. Despite this, most students found processing this data a challenge, with less than 5% of responses being awarded full marks. Many students did not refer to the standard deviations in their answers and when they did, it was often in the context of describing the degree of variation in the results for that particular treatment. The students who did notice the overlapping standard deviations between treatments A and C often referred to the results being significant rather than stating that the difference between these groups is likely to be significant.

Over 75% of students gained at least one mark in **03.4** and this was typically for describing how aphids feed by inserting the stylet into the phloem. Although descriptions of the stylets were allowed, it was good to see that many students did refer to the stylets by name in their answers. In some cases, there were vague references to the stylets being inserted into the stem or the leaf. Fewer students considered the ideas behind the third marking point eg aphids as vectors of plant pathogens or what the consequences would be for the plant if there are fewer sugars or amino acids.

QUESTION 04

This question was set on the familiar context of using a potometer to measure rate of water uptake. Despite this, **04.1** proved to be a good discriminator with less than 5% gaining all three marks and around 25% gaining two marks. Many answers correctly referred to cutting the shoot underwater or assembling the equipment underwater for the first marking point. Fewer students were able to describe how to prevent leaks from the potometer or how to prevent the air bubble from coming off the scale.

Only around 30% of students achieved the full three marks in question **04.2** There were many different errors observed, such as not using πr^2 or using diameter rather than radius. It was common for students to use incorrect values from the graph, or either forget to divide by 30, or divide the volume by the distance.

Students found question **04.3** a challenge with around 40% gaining at least one mark and less than 5% gaining the full three marks. Many students were not able to apply knowledge of water movement into the guard cells by osmosis to explain that the stomata would open. Very few students could therefore link the higher light intensity to more stomata being open resulting in the higher water uptake shown in Figure 4.

QUESTION 05

Around 50% of students gained the mark in **05.1** and correctly suggested that the artery appears as circular due to the thicker wall. Some answers incorrectly described the vein as having the thicker wall or that the artery appearing as circular was just due to chance.

In **05.2**, over 50% of students gained the mark. Where the mark was not awarded, it was usually for not using information from Table 2 or for not giving the answer as a word equation as instructed. Over 10% of students did not attempt the question.

The calculation in **05.3** was answered well, with 75% of students gaining the mark. The most common errors observed were either not giving the answer to the nearest whole number or expressing the ratio the other way round.

In question **05.4**, 50% of students correctly applied knowledge of capillaries to suggest the advantage of the low blood flow. Where marks were not awarded, it was typically for not referring to time to allow for exchange. There were several incorrect suggestions, such as the low rate of blood flow being important so that capillaries do not burst.

Question **05.5** was answered well with close to 60% gaining full marks and around 30% achieving one mark. Where students missed marks, it was usually for missing one aspect or for drawing multiple lines as bars to represent rates of flow.

Close to 80% achieved the mark in **05.6** and correctly described the relationship between mean diameter and rate of blood flow. Occasionally, students incorrectly referred to a negative correlation.

QUESTION 06

Question **06.1** was not answered very well, with only 40% of students gaining the one mark. Many just referred to the antigens as markers or proteins on cells, without any reference to stimulating an immune response. Some answers were closer, eg antigens bind to immune cells but again without reference to the following immune response.

In question **06.2** students had to describe stage 1 and stage 2 from Figure 6. Whilst 70% achieved at least one mark, only around 8% achieved full marks. The process of the B cell presenting the pathogen antigens on the surface was commonly missed. Some answers confused the B cell for a phagocyte and therefore did not gain the first marking point.

Around 50% correctly named the memory cells in question **06.3** but only a further 30% went a stage further to correctly explain the function. There were many vague answers, eg to memorise the pathogen for future immune responses or to retain information about the pathogen/antigen. Some just simply stated to provide long-lasting immunity.

There was a great deal of variety in the answers to **06.4** with many diagrams not being drawn clearly or without any labels. Just over 70% of students gained at least one mark and only around 8% of students achieved full marks. The most common mark awarded was for diagrams showing two heavy and two light chains. It was surprising that not many diagrams were labelled with the antigen binding sites and some were labelled as active sites. Very few students included labels to disulfide bridges.

Just over 50% gained the full two marks for the calculation in **06.5** and just over 25% gained one mark. Where marks were lost, it was usually for either not giving the answer in standard form or for answers not given to 2 significant figures. Some students had difficulties converting 48 hours into seconds before multiplying by 2000.

Question **06.6** was answered well, with over 75% of students gaining at least two marks and around 40% achieving full marks. Where students did not gain certain marks, it was usually for not giving the idea of ‘many’ at least once within the answer or for incorrectly naming certain organelles. Some students named ribosomes and rough endoplasmic reticulum, but this was under the same marking point. Overall, fewer students referred to the Golgi body in the role of modification and/or transport antibodies out of the cell.

QUESTION 07

Questions **07.1** and **07.2** were not answered particularly well, despite both questions consisting of purely AO1 marks.

In **07.1**, just under 50% achieved at least one mark, around 25% achieved at least three marks and just over 5% achieved full marks. A good number of students included details of how lipids are digested, which was not required for this question. The majority of responses lacked many details, such as including the endoplasmic reticulum or the role of the Golgi body. There were also many responses that included chylomicrons moving directly into capillaries, rather than into the lacteals. The role of micelles was generally misunderstood by many students.

The general performance in **07.2** was similar to that of **07.1** but even fewer students achieved full marks (less than 5%). Many answers had the general ideas for a good response but lacked specific detail and/or were missing certain keywords. There was misunderstanding in how and where atheromas form, with many responses just stating, ‘in the lumen of the artery’ or ‘on the artery lining’. Many answers did not name the coronary arteries at any point or what the consequences of reduced blood flow to the heart muscle tissue would be.

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